

Nakul Iyer

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EDUCATION

Purdue University

B.S. in Computer Engineering

West Lafayette, IN

Aug. 2025 – May 2028

- **Relevant Coursework:** Multivariate Calculus, Ordinary Differential Equations, Discrete Mathematics for Computer Engineering, Electrical Engineering Fundamentals I/II, Advanced C Programming

EXPERIENCE

Design Lead | *Purdue Engineering Projects in Community Service*

Jan. 2026 — Present

- Lead a team of 8 students to develop a gamified early literacy application for the Boys & Girls Club of Northern Indiana Corridor, aligning structured learning modules and UI workflows with the UFLI phonics curriculum to support classroom and at-home literacy development.
- Engineer core application features using the Flutter + Firebase stack, including sound identification exercises, CVC word construction, guided tracing interfaces, diagnostic placement logic, and level-based reward progression; implemented scalable backend data models to track student progress and support adaptive learning pathways.

Software Developer | *Hack the Future*

Oct. 2025 — Present

- Built a centralized internal operations platform for Sleep Pod, a UK-based nonprofit distributing insulated sleeping pods across the UK and Europe, using the MERN stack (MongoDB, Express.js, React, Node.js) to replace fragmented WhatsApp, email, and spreadsheet workflows.
- Designed and implemented an analytics module using Apache ECharts with dual-party data verification, automated system-generated events for prison partners, and interactive visualizations for production amounts, material breakdowns, and geographic distribution.
- Built a real-time pending/history event pipeline using React, Node.js, Express.js, MongoDB, Stream Chat API, FullCalendar, and Apache ECharts, with role-based card views, dispute detection, admin override capabilities, and filtering by date, user, organization, and item type.

Research Intern | *Weill Cornell Medicine*

Aug. 2022 — Apr. 2024

- Contributed to AutoFocus, an R-based hierarchical clustering model that organizes multi-omic biomolecules into correlation-based trees and tests phenotype enrichment across every internal node.
- Built interactive R/Shiny functionality for exploring enriched disease-associated modules, functional annotations, and phenotype drivers across biomolecular hierarchies.
- Applied the framework to ROS/MAP Alzheimer's brain metabolomics and proteomics data, identifying multi-scale modules associated with neurofibrillary tangles and cognitive decline.

PROJECTS

Machine Learning Image Classification Model | *Python, NumPy, pandas, Matplotlib*

Nov. 2025 — Present

- Implemented multiple machine learning classifiers from scratch, including K-Nearest Neighbors, Decision Trees using information gain, and Logistic Regression, to classify images based on extracted features.
- Designed data preprocessing pipelines involving feature standardization, train/validation/test splits, and hyperparameter tuning to optimize model performance.

Gem Hunters Game | *Python, Javascript, JsPsych, Node.js*

June 2024 — Aug. 2024

- Built a web-based multidimensional reinforcement learning task in JavaScript using jsPsych where participants completed 216 trials each to learn reward rules that were changed between rounds.
- Engineered randomized experiment logic and data collection pipelines, implementing probabilistic reward schedules (80/20) and reaction time logging across 20,000+ trials, enabling precise measurement of learning performance.
- Deployed the experiment to 100+ participants via Prolific and analyzed response patterns, producing statistically usable data that was presented to 40+ professors, PhD students, and graduate researchers at the Mount Sinai.

TECHNICAL SKILLS

Languages: R, Python, Java, JavaScript, HTML/CSS, L^AT_EX, React, SQL, MATLAB

Libraries: Tidyverse, ggplot, jsPsych, NumPy, scikit-learn, Matplotlib, pandas, Node.js, Express.js

Tools: GitHub, Cursor, Visual Studio Code, MongoDB, Docker, Claude Console, Jira